

Foto: Dr. Matthias Otte

This work has been carried out within the framework of the EUROfusion Consortium and has received funding from the Euratom research and training programme 2014–2018 under grant agreement No. 633053. The views and opinions expressed herein do not necessarily reflect those of the European Commission.



EUROfusion

HELMHOLTZ

RESEARCH FOR GRAND CHALLENGES



IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

March 14, 2026

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MOTIVATION

NUCLEAR FUSION: ENERGY CHALLENGE

- Lawson criterion for net energy gain:

$$n_e T \tau_E \geq \frac{12 f_{\text{tot}}}{\langle \sigma_{\text{DT}} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$

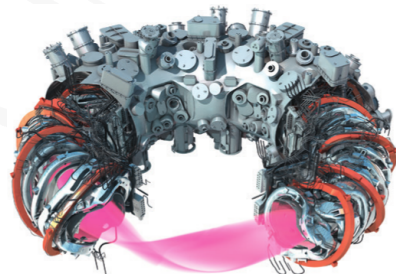
- Requires: high temperature ($T \sim 10 \text{ keV}$ to 20 keV), density, and confinement time

Triple Product

Key figure of merit for fusion performance combining density, temperature, and energy confinement time

WENDELSTEIN 7-X

- World's largest stellarator (major radius 5.5 m, minor radius 0.53 m)
- 70 superconducting magnets: 50 non-planar & 20 planar coils
- Volume 30 m^3 , $B_{\text{max}} = 3 \text{ T}$
- Island divertor configuration (5/6, 5/5, 5/4)
- Optimized for reduced neoclassical transport
- Heating power up to 14 MW ECRH



Key Challenge

Managing heat loads on plasma-facing components during long, high performance plasma

PLASMA RADIATION AND TRANSPORT

Transport Mechanisms:

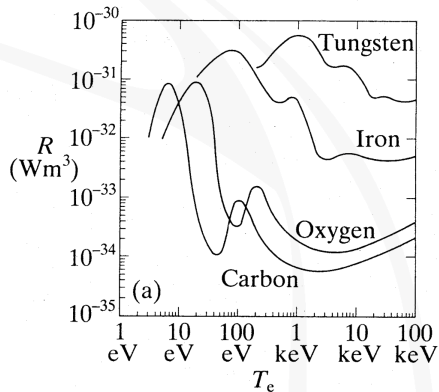
- Classical: collisional diffusion
- Neoclassical: trapped particles, $\propto 1/\nu$
- Anomalous: turbulence (dominant)

Radiation:

- Bremsstrahlung: $P_{\text{Brems}} \propto n_e^2 T^{1/2}$
- Line radiation: atomic transitions
- Impurity effects on plasma performance

Impurity Seeding

Injection of low-Z (He, N₂) or high-Z (Ne, Ar) impurities to enhance edge radiation and cooling

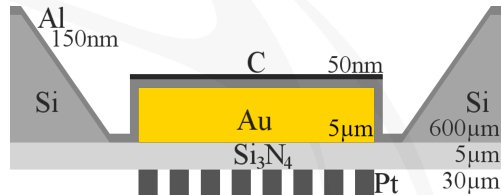


BOLOMETER DIAGNOSTIC

BOLOMETRY AT W7-X

Metal Resistor Bolometers:

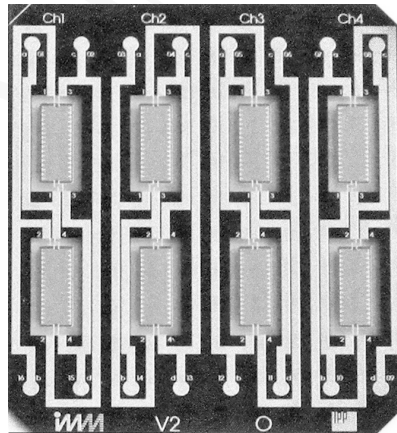
- Thin metal (gold) film absorbers
- Wheatstone bridge circuit configuration
- Measures total radiation power (broadband)
- Multicamera system: HBC & VBCI/r (32 + 2×24 channels)



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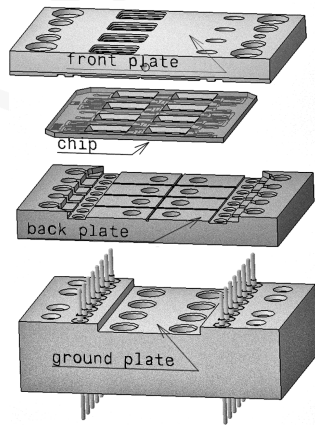
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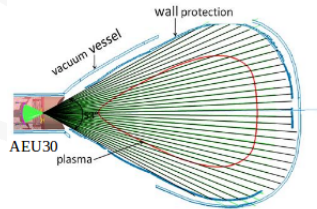
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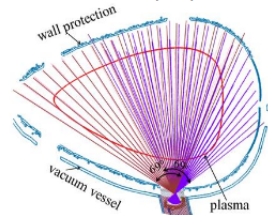
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horizontal bolometer camera (HBC)

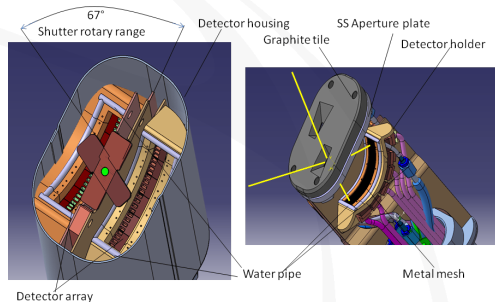


vertical bolometer camera (VBC)

BOLOMETRY AT W7-X

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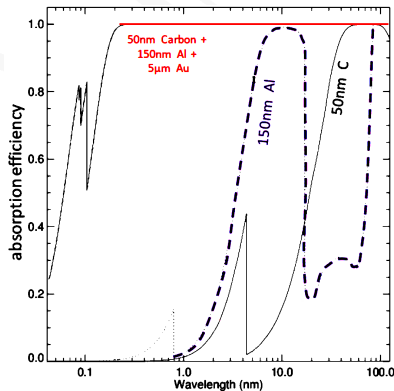
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PERFORMANCE

Performance Characteristics:

- In-situ ohmic heating calibration (F_M , f_M)
- Temporal resolution: $\{0.4, 0.8, 1.6, \dots, 6.4\}$ ms
- Measurement uncertainty (*Gauss*): $1.6 \mu\text{W} \leftrightarrow \text{SNR} > 1000$
- Spatial resolution: ~ 5 cm at magnetic axis
- Near unity absorption efficiency: 600-0.2 nm



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Power Per Detector:

$$P_M = F_M \cdot \left(\frac{d(\Delta \tilde{U}_M)}{dt} + f_M \Delta \tilde{U}_M \right)$$

Global Radiation Power:

$$P_{\text{rad}} = \frac{V_{\text{P,tor}}}{V_C} \sum_M \frac{P_M V_M}{K_M}$$

PERFORMANCE

Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information ($P_{\text{chord}} \sim P_M/K_M$)

Applications:

- Power balance calculations
- **Tomographic reconstruction**
- **Real-time feedback control**

→ **controlled radiative cooling in the plasma edge**

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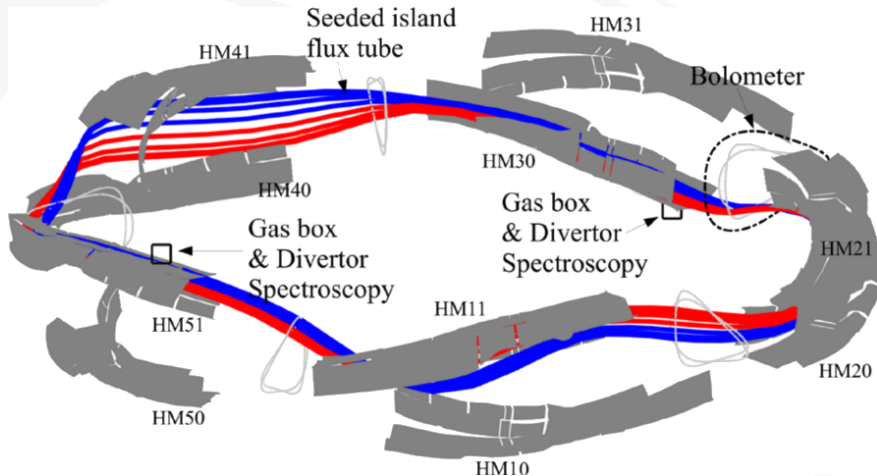
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REAL-TIME RADIATION FEED- BACK CONTROL

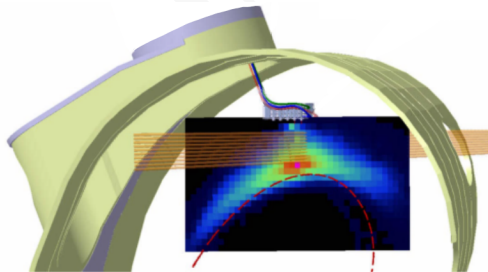
FEEDBACK SYSTEM



FEEDBACK SYSTEM

Architecture:

- Integrated data acquisition hardware & optimized control software
- Twin proxies and dedicated signal outputs
- Thermal gas valve actuation with multipurpose/-diagnostic PID controller



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Radiation Prediction Proxies:

A: *Channel selection S*

$$P_{\text{pred}}^{(1)} = \frac{V_{\text{P,tor}}}{V_S} \sum_M^S \frac{P_M V_M}{K_M}$$

(e.g. $S_{\text{VBC}} = \{61, 86, 78, 73, 70, 65\}$)

B: *Single channel*

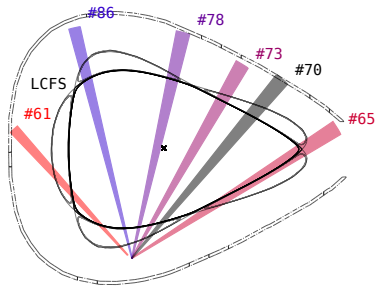
$$P_{\text{pred}}^{(2)} \propto \Delta \widetilde{U}_M$$

(dimensionless, minimal overhead)

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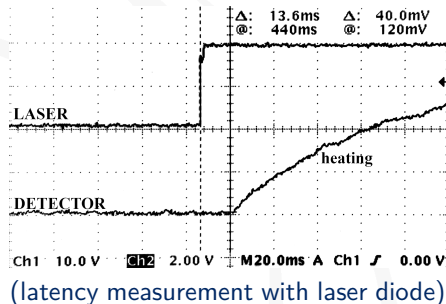
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FEEDBACK SYSTEM

System Performance:

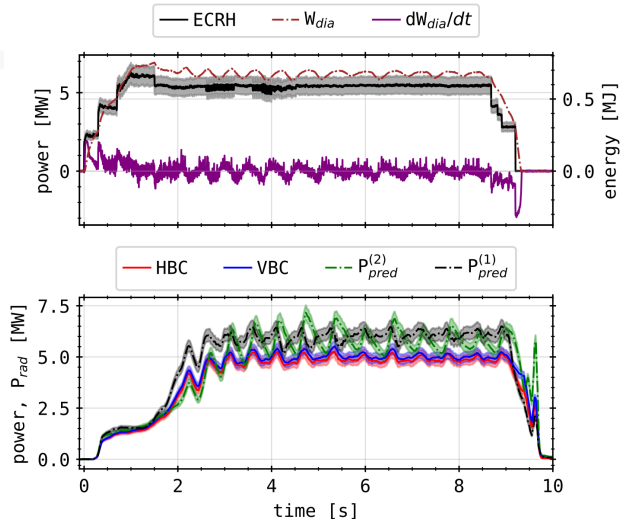
- FIFO smoothing in $P_{\text{pred}}^{(1)}$: $\sim M \cdot \Delta t / 2$ ($M=10$, $\Delta t=1.6$ ms)
- Minimum acquisition latency 13.6 ms
- Total system latency: 150 ms to 400 ms (dominated by plasma response)



EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

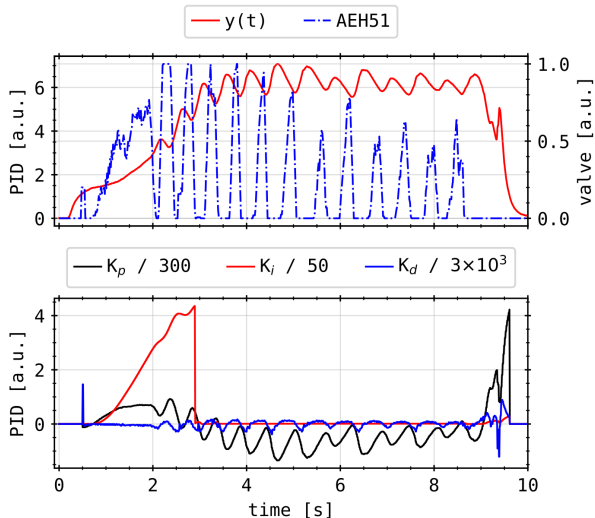
- Duration 9.2 s
 $P_{\text{ECRH}}^{\text{max}} = 6.23 \text{ MW}$ (O2-mode)
- Total input energy 55 MJ
- Radiation fraction PID
 setpoint $f_{\text{rad}} \sim 90\%$
- 5-channel VBC subset
 $S = \{65, 78, 61, 73, 86\}$



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EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

- Peak f_{rad} up to 100%
- Target heat load reduction: factor >2
- C^{3+} detachment visible at $f_{\text{rad}} \sim 50\%$

Key Result

Demonstrated stable, feedback-controlled radiative cooling with $f_{\text{rad}} \geq 85\%$ without terminal plasma disruption

EXPERIMENTAL ACHIEVEMENTS

Comparison with Other Feedback Methods:

Electron Density Feedback:

- Indirect control of radiation
- Density affects confinement
- Less direct connection to detachment

C-III Filterscope Feedback:

- Spectroscopic proxy
- Single impurity species
- Limited spatial information

Bolometer Feedback:

- Direct radiation measurement
- Intrinsic connection to detachment physics
- Total power control
- Multiple viewing chords
- Robust to impurity mix changes

Advantage

Bolometer feedback directly controls the relevant quantity (radiation) unlike density or spectroscopy proxies